DOCUMENT 00 · HYBRID HOST ENGINEERING BASELINE

MCRS HYBRID

Modular Captured Ride System · CMFSB

Captured Mechanical Floating Safety Body

Two hybrid (ICE + e-drive) reference hosts · V-L (4-tower Class S) · V-H (4-tower Class L) · COMFORT / SPORT captured upper body

Field	Controlled value
Document ID	MCRS-DOC-00-R5.40
Issue date	2026-10-12
Founder / Legal Owner	Amirbahador Atai (individual rights holder; assignment to a legal entity is a separate controlled act)
Licensable module	MCRS-MOD-001 (four- or six-tower CMFSB isolation module)
Classification	T1 Confidential / NDA controlled
Document class	Hybrid-host baseline (module + hybrid reference hosts); module values governed by Master MCRS-DOC-00-R8.09
Supersedes	MCRS-DOC-00-R5.39. Review R5.40 (RX-01 to RX-06) applied: G1 / G2 exit criteria include REQ-H07, REQ-H08 and REQ-M32 to REQ-M36; release readiness and contract use (Appendix E); owner decisions D-01 to D-08 unchanged.
Status	Proposed baseline, pending owner release and independent technical review. Evidence level L0, TRL 2–3.
Independent Technical Reviewer	To be appointed no later than the earlier of 2026-12-05 and G1 closure, and before any external safety-performance credit.
Approval authority	TBD after independent review

This document is a design baseline and diligence instrument. It does not certify range, crashworthiness, homologation, ASIL, production readiness, patentability or freedom to operate. OPEN items remain OPEN until evidence closes them. Module values are controlled exclusively by Master MCRS-DOC-00-R8.09; this document controls host values only.

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1. Purpose, scope and positioning

This baseline defines the integration of the CMFSB isolation module (MCRS-MOD-001) on two hybrid reference hosts (ICE + e-drive): V-L (4×4, four seats) and V-H (6×6, six seats). It is the hybrid-host controlling Document 00 under the module-centric Master MCRS-DOC-00-R8.09. The Master governs all module values; this document governs hybrid-host values and mirrors only the module sections needed for standalone readability.

Document	Role	Governs
MCRS-DOC-00-R8.09	Module-centric Master	Module values; cross-host comparison
MCRS-DOC-00-R5.40 (this file)	Hybrid-host baseline	Hybrid-host values only
MCRS-FE-DOC-00-R7.09	Electric-host baseline	Electric-host values (subordinate to Master for module)

> One value, one place. Module sections here are mirrors by citation; the Master governs. Host values live in §5–§8 and §13. Differences between hybrid and electric hosts are in §2.3.

1.1 Value statement and product framing

Occupant compartments in specialty, defence-utility, ambulance and limited-series vehicles receive road vibration that degrades comfort, task performance and, where relevant, clinical stability. Soft isolation alone can reduce handling stiffness in rapid manoeuvres. CMFSB addresses both: the upper body stays mechanically captured at all times, with a COMFORT isolation envelope, a SPORT mechanical lock and passive support on power loss. Reference application RA-1 (ambulance / specialty medical transport) is the context for comfort requirements; no clinical outcome is claimed.

State	Experience	Architecture behaviour	ID
COMFORT	Road harshness isolated from the cabin	Locks released; macro and micro stages active; leveling active	S2
SPORT	Body feels planted in rapid manoeuvres	All required locks confirmed; isolation stages held	S2S
Power loss	Cabin settles on a known mechanical path	Powered valves fail-closed; passive support; no software required	S4

1.2 What is offered

- Path A (primary): licence of the module under a paid Option (SIM-001 run and ICD-001 draft); the buyer may bring its own host.
- Path B: host engagement studies using the hybrid reference host; no vehicle sale is implied.
- Path C: IP-transaction discussion under counsel (T3 only).

2. Executive summary

The CMFSB module places a captured, isolated upper safety body on four or six capture towers. Safety rests on mechanical capture, passive support and stops, never on the actuator or on software. The hybrid hosts supply an ICE + e-drive powertrain, a 48 V / 12 V and small HV electrical system, a fuel system and the mass budget against which the module is evaluated.

Hybrid host item	V-L	V-H	Status
Architecture	Hybrid 4×4, four towers (Class S)	Hybrid 6×6, four towers (Class L)	BASELINE
ICE class	V8, 4.0 L twin-turbo class	V12, 6.0 L twin-turbo class	TARGET
Electric traction / battery	≈150–180 kW / 20 kWh usable	≈200–250 kW / 40 kWh usable	TARGET
Fuel tank / range target	80 L / 1,000 km (cycle ≤70 km/h avg)	150 L / 700–800 km	STUDY
Overall height (SPORT)	1,950 mm	2,130 mm	FIXED-FOR-ENGAGEMENT
Curb / design GVW	2,778 / 3,278 kg	4,983 / 5,983 kg	TARGET
GVW ceiling / margin	4,200 kg / 922 kg	7,000 kg / 1,017 kg	LOCKED / DERIVED
Stack residual	0 mm (conditional)	+10 mm (conditional)	OPEN — P0
Evidence level	L0	L0	BASELINE

2.1 Readiness assessment (hybrid host)

Dimension	Position	Principal exposure	Closure vehicle
Architecture	Defined and consistent at L0	No hardware or simulation evidence	SIM-001 (G1), tower bench (G2)
Range	Fuel-range screening with efficiency band (§7)	V-L 1,000 km holds only to ≈74 km/h average at η 0.22; CdA scaled with height	ENE-H-001, fuel maps, CFD
Mass	Closed at P50 with explicit reserve	Four-corner Class L P90 427–565 kg vs 393 kg budget; Class S exposure +113 kg is covered by the 113 kg V-L reserve (D-08); evidence still required	G2 supplier data, G1 Class S study
Packaging	Re-baselined stack closes at 0 / +10 mm	Depends on 170 mm of unvalidated levers; zero margin on V-L	G1 3-D datum stack
Safety	Invariants and screening FMEA	Fuel, heat and engine excitation near the cell	SAF-001, THERM-001 (G2)
Regulatory	ICE market exposure not assessed; V-L ceiling 4,200 kg is above the 3,500 kg threshold in some jurisdictions (design GVW 3,278 kg is below it)	Restrictions vary by region	REG-001 (G1)
IP	Ownership chain and filing OPEN	No filing receipt on record	Counsel, provisional filing

2.2 Revision history

Revision	Scope
R5.32	Combined module and hybrid-host baseline
R5.33	Hybrid host separated from module content; fuel-range screening; Class L sizing; ICE class TARGET
R5.34	Two content passes (Corrected, Fixed) under one ID: mass sums, P90 four-tower 427 / 565 kg, aux load in fuel screening, cycle definition, lock / S4 clarification, status tags
R5.35	New ID. Owner decisions D-01 ceiling 4,200 kg, D-02 four V-H towers, D-03 heights 1,950 / 2,130 mm; stack re-baselined; fuel range recomputed with CdA scaled to height (V-L cycle ≤70 km/h); P90 method stated exactly; Class S exposure; Master reference R8.04
R5.36	New ID. Owner decision D-05: body zoning and load boundary (§3.4, mirrored from Master R8.05): only the passenger cell is suspended; battery, powertrain and fuel on the lower chassis; payload-location sensitivity; seam envelope; Master reference R8.05
R5.37	New ID. Owner decisions D-06 (electric V-L battery 275 kWh in the comparison) and D-07: only the safety cockpit is suspended; hood, trunk and payload on the lower body. Upper sprung mass recomputed without payload (V-L ≈1,155 kg; V-H ≈2,020 kg with protection); 20 g corner ≈57 / 99 kN; Master reference R8.06
R5.38	Review of R5.37: consistency items EC-01, EC-05, EC-07, EC-09, EC-10; engineering findings EN-01 to EN-06; REQ-H07 and REQ-H08; Master reference R8.07
R5.39	New ID. D-08: V-L reserve 113 kg (curb 2,778 kg, GVW 3,278 kg, margin 922 kg); fuel range recomputed with reference mass 2,978 kg; stop-stroke efficiency and design speed, horizontal capture bound, seam envelope and height sensitivity (Appendix D); page count above the previous issue; Master reference R8.08
R5.40 (this issue)	New ID. Review RX-01 to RX-06: G1 / G2 exit criteria include the new requirements; assumption register and release-readiness statement (Appendix E); Master reference R8.09

2.3 Reference host comparison

The electric V-H design GVW equals the 7,000 kg ceiling, so payload is the balancing item; the hybrid hosts carry more margin. Both V-H hosts now use four Class L towers (D-02).

Item	V-L electric	V-L hybrid	V-H electric	V-H hybrid
Powertrain	BEV, dual e-axle AWD	ICE V8 4.0 L TT + e-drive	BEV 6×6, three e-axles	ICE V12 6.0 L TT + e-drive
On-board energy	275 kWh usable	80 L + 20 kWh	300 kWh usable	150 L + 40 kWh
Range target (STUDY)	≥800 km	1,000 km (≤70 km/h avg)	≥800 km utility (≤≈70 km/h)	700–800 km
Towers	4 (Class S)	4 (Class S)	4 (Class L)	4 (Class L)
Curb target	3,544 kg	2,778 kg	5,695 kg	4,983 kg
Design GVW	4,044 kg	3,278 kg	7,000 kg	5,983 kg
Margin to ceiling (4,200 / 7,000 kg)	156 kg	922 kg	0 kg	1,017 kg
Stack residual	0 mm	0 mm	+10 mm	+10 mm
Dominant host risk	Pack mass; ceiling margin	ICE range claim, fuel and heat	Pack mass; 800 km only at low speed	Four-corner P90 mass, fuel and heat
Evidence	L0	L0	L0	L0

3. Product definition and architecture

The host is defined around the CMFSB load and isolation architecture; it is not a powertrain retrofit. The MCRS tower set is the sole mechanical connection between the two body pieces in normal operation. Active isolation works inside a bounded mechanical safety envelope; comfort control, handling stiffness and crash containment are coordinated but independently bounded.

3.1 Layer stack (top to bottom)

Layer	Function	Safety role
Upper safety cell	Occupant and mission body (CFRP, roll structure, seats, closures)	Captured at all times
Micro stage (Lorentz)	High-bandwidth isolation within ±7.5 mm	Force-bounded; never a crash or horizontal path
Macro stage (hydraulic)	Leveling, ±50 mm body-relative travel, lock seats	Maintains geometry and ride envelope
Progressive stops	Energy absorption on abnormal vertical events	Primary abnormal vertical energy path
SPORT positive lock	Pin, claw or hydraulic clamp at the seat band	Handling stiffness and load sharing; not a replacement for stops
Capture frame and links	Longitudinal, lateral and yaw restraint	Primary horizontal path
Lower chassis hardpoints	Load transfer into the host platform	Structural continuity

Figure 1 (conceptual). Two-piece body: the upper cell is captured on MCRS towers above the lower host chassis. V-L: four towers (Class S). V-H: four towers on both hosts (owner decision D-02); six towers remain an option of MCRS-MOD-001. Vertical support is passive macro support; horizontal loads go through the capture frame and links; abnormal vertical energy goes through progressive stops.

3.2 Independent load and force paths

Load / function	Primary path	Secondary / containment
Static vertical support	Passive hydro-pneumatic macro support	Progressive stops / chassis structure
COMFORT dynamic force	Lorentz micro actuator (force-bounded)	Macro stage + passive support
Large displacement, leveling	Hydraulic macro stage	Stops
SPORT stiffness	Positive locks + seat faces	Capture frame
Longitudinal / lateral / yaw	Capture frame, links, horizontal stops	SPORT lock faces
Crash / abnormal vertical	Progressive stops + structural hardpoints	Capture frame / links
Power loss	Passive hydraulic / compliant support (S4)	Mechanical capture
Battery crash load	Lower chassis + protected HV enclosure	Independent containment structures

The Lorentz stage is an actuator option. If another actuator family replaces it, the safety invariants and interface requirements still apply under configuration control. Hold is an optional, rate-limited secondary restraint (orifice and/or friction at the macro stage); it is distinct from the SPORT lock and the stops and is not a primary crash path.

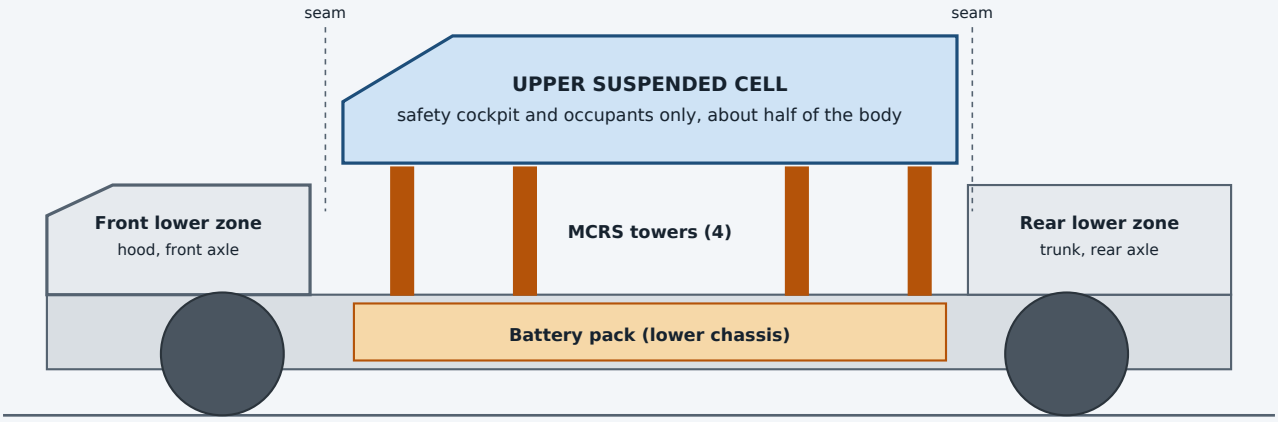
3.3 Qualitative comparison

Approach	Comfort	Handling stiffness	Positive lock	Independent capture	Power loss
Soft air / hydro-pneumatic	Good	Reduced in rapid manoeuvres	No	Limited to suspension travel and bump stops	Settles on bump stops
Adaptive dampers	Improved trade-off	Improved	No	No	Fixed or fail-soft damping
CMFSB / MCRS (target)	Targeted (unproven at L0)	Targeted via SPORT lock	Yes (SPORT)	Yes: stops, frame, links	Settles on passive support, captured

Qualitative only; no measured superiority is claimed at evidence level L0.

3.4 Body zoning and load boundary (owner decisions D-05, D-07)

Only the safety cockpit is suspended: the passenger cell with the occupants' space, about half of the overall body (planning ≈2,500 mm of the 5,000 mm V-L body and ≈2,750 mm of the 5,500 mm V-H body). The hood, front wheel housings, trunk, rear body and wheel housings are attached to the lower body, and the lower chassis forms a cradle around the cell. The battery, e-drive or ICE, fuel tank, axles, brakes, steering and the trunk payload are all on the lower chassis.



Battery, powertrain, fuel, hood and trunk payload weigh on the lower chassis only. The module carries only the cockpit.

Figure 1b (conceptual, not to scale). Side view of the zoning: the upper cell rides on the MCRS towers; battery, powertrain and axles belong to the lower chassis.

Zone	Contents	Carried by	Loads the module?
1 Upper suspended cell (safety cockpit)	CFRP safety cell, roll structure, seats, closures, occupants (no payload, no battery)	MCRS towers	Yes: this is the upper sprung mass
2 Front lower zone	Hood, front wheel housings, front axle, front crash structure	Lower chassis	No
3 Rear lower zone	Trunk, rear body and wheel housings, rear axle(s), payload	Lower chassis	No
4 Lower chassis	Battery pack (electric V-L 275 kWh / V-H 300 kWh; hybrid 20 / 40 kWh), e-drive / ICE, fuel tank, brakes, steering, thermal system	Wheels and suspension	No: never loads the module

How the battery relates to the upper level. It adds no load to the module or the cell. It matters only through (a) total vehicle weight against the GVW ceiling and (b) its thickness inside the lower chassis, which sets the chassis height in the packaging stack (pack volume ≈1,020–1,665 litres). The tower depth shared with the chassis and battery floor (60 mm lever) is the only place where battery packaging and module packaging overlap.

Seams. The joints between the moving cell and the fixed front and rear bodywork shall accommodate ±57.5 mm of normal relative motion (50 mm macro + 7.5 mm micro) plus the stop stroke (50 mm placeholder) without contact, using flexible seals or bellows; values are frozen in ICD-001. Seam water, dust, noise and pinch protection is not yet designed (OPEN).

Payload location (decision D-07). Payload is carried in the trunk on the lower body and does not load the module; the upper sprung mass is the cockpit structure plus occupants (and the optional protection package where fitted). Had payload been carried in the cabin, the upper mass would rise by 100 kg (V-L) and 705 kg (V-H electric), and the V-H electric 20 g corner load from ≈101 to ≈135 kN. Trunk payload loads the lower chassis and rear axle instead (STR-001, G1).

4. Safety invariants and safety layers

§4 is the controlling statement of the safety invariants. Appendix I and both host baselines mirror it; the mirrors are labelled and §4 prevails on any difference.

Invariant	Rule
Mechanical capture	No operating state permits loss of geometric capture; no free levitation.
Static support	Vehicle weight on passive or fail-safe compliant paths, never the sole magnetic path.
Power loss (S4)	Passive support; powered leveling-supply and hold valves fail-closed; the hydro-pneumatic spring path (cylinder to accumulator) stays connected; software aliveness not required.
Crash path	Progressive stops + structure, independent of closed-loop magnetic force.
Mode transition	SPORT↔COMFORT never enters unconstrained free-float.
SPORT lock fault	Lock failure shall not defeat capture or progressive stops; SPORT is never declared with an incomplete lock set.
Horizontal restraint	Longitudinal / lateral / yaw loads resolved by dedicated paths and/or SPORT lock.
Override limit	Driver override cannot defeat the lateral-g interlock or a critical fault state.
Manual rescue	Two-step rescue available under power loss and hydraulic failure.
HV isolation	Traction HV galvanically isolated from the module control domain; a single HV fault shall not defeat capture or S4.
Thermal / fire	Battery runaway containment is a lower-chassis responsibility and shall not propagate into upper-body capture integrity.
Single-path rule	No single active-control path is credited as the primary crash path.
Lock state on power loss	Locks remain engaged only if already engaged at the instant of loss (SPORT). From COMFORT the locks stay released. Logic frozen in ICD-001 / LCK-001 (REQ-M17 / M20).

4.1 Layered safety response

Safety layer	Failure addressed	Design response
Mechanical capture	Link / carriage failure	Capture frame with redundant restraint paths
Passive support	Power loss	Hydro-pneumatic / compliant support, fail-safe valves
Progressive stops	High-energy vertical event	Defined force–displacement curve (STR-001)
Horizontal restraint	Braking, cornering, crash	Dedicated links, stops and lock faces
Positive SPORT locks	Mode stiffness, lock integrity	Per-lock confirmation; incomplete set inhibits SPORT
HV safety	Electrical crash, isolation fault	HVIL, isolation monitor, service disconnect, crash inhibit
Thermal safety	Battery / inverter overheating	Redundant sensing, derating, propagation-resistant pack, vent routing
Control safety	Bad command, software fault	Bounds, diagnostics, degraded state, independent monitor

No ASIL assignment is claimed. High-severity (S ≥ 9) items require explicit pre-G2 review independent of RPN score.

Mirrored from Master §4 and §4.1; the Master governs on any difference.

5. Host definition and packaging gate

5.1 Geometry and mission

Parameter	V-L hybrid	V-H hybrid
Length / width (mirrors excl.)	5,000 / 2,000 mm	5,500 / 2,200 mm
Overall height (SPORT)	1,950 mm (re-baselined)	2,130 mm (re-baselined)
Wheelbase	3,000 mm	3,300 mm (front axle to mid of rear tandem)
SPORT ground clearance	200 mm	300 mm
Drive	4×4	6×6 (front + dual rear tandem)
Seats	4	6
Passenger cell	Full CFRP structural cell, budget ≤280 kg	Full CFRP structural cell, budget ≤420 kg
Roll structure	FIA-referenced CFRP / hybrid, ≤45 kg (no homologation claim)	Same, ≤70 kg
MCRS class / towers	Class S / 4	Class L / 4 (decided; six-tower stays a module option)
GVW engagement ceiling	4,200 kg	7,000 kg
Optional light protection package	—	≈200 kg; treated identically in both hosts

Heights, ground clearance and cabin definition are engagement targets read as FIXED-FOR-ENGAGEMENT / PACKAGING GATE OPEN. Status tags are defined in Appendix B.

V-H tower count (decided). Four Class L towers on both V-H hosts (D-02). The four-corner Class L P90 is 427 kg (RSS), 439 kg (pump correlated) and 565 kg (arithmetic) against a 393 kg budget (+34 / +46 / +172 kg); the 150 kg reserve covers both RSS cases only. The six-tower option would lower the 20 g load per tower from ≈99 to ≈66 kN but add ≈172 kg; it is not baselined.

5.2 Vertical packaging stack (P0 open item)

Element (mm)	V-L	V-H	Status
Overall height (SPORT)	1,950	2,130	FIXED-FOR-ENGAGEMENT
Ground clearance (SPORT)	200	300	FIXED-FOR-ENGAGEMENT
Available for structure (height − GC)	1,750	1,830	DERIVED
Chassis / axle structure	280	320	STUDY
Tower structure + mounts (60 mm shared with chassis)	60	80	STUDY
Macro travel (100) + free play (10)	110	110	STUDY
Stop stroke + micro-stage clearance (placeholder)	50	50	ASSUMPTION
Floor-to-H-point	220	220	STUDY
Cabin, H-point to roof	960	960	TARGET
Roof / exterior shell	70	80	STUDY
Allocated sum	1,750	1,820	DERIVED
Residual margin	0	+10	OPEN — P0 (conditional)

Closure lever adopted (mm recovered)	V-L	V-H	Basis
Share tower depth with chassis / battery floor (120 → 60 / 140 → 80)	60	60	Integration study
Reduce travel + free play (140 → 110; free play 40 → 10)	30	30	Stop-stroke trade
Cabin H-point to roof 1,000 → 960	40	40	Ergonomic package
Floor-to-H-point 250 → 220	30	30	Seat / floor design
Shell thickness (80 → 70 / 90 → 80)	10	10	CFRP roof design
Total recovered	170	170	

Heights were raised from 1,700 / 2,000 mm to 1,950 / 2,130 mm (owner decision D-03). Residual 0 mm (V-L) and +10 mm (V-H) holds only if every lever is delivered; none is validated, V-L has zero margin, and the 50 mm stop-stroke allowance is a placeholder until STR-001. Status stays OPEN — P0 until the G1 3-D datum stack confirms it.

6. Powertrain and fuel system

6.1 Powertrain

Powertrain	V-L hybrid	V-H hybrid	Status
Architecture	Hybrid (ICE + e-drive)	Hybrid (ICE + e-drive)	Engagement target
ICE class	V8, 4.0 L twin-turbo class	V12, 6.0 L twin-turbo class	Engagement target
ICE power	≈430–450 kW class	≈460–500 kW class	TARGET
Electric traction	≈150–180 kW total	≈200–250 kW total	TARGET
Battery usable energy	20 kWh	40 kWh	Engagement target
Fuel tank	80 L	150 L	Engagement target
Combined range target	1,000 km (cycle avg ≤70 km/h)	700–800 km	STUDY

Cell chemistry, ICE supplier, transmission and e-drive suppliers are not frozen in Document 00; they belong to ICD-001 and supplier releases. Fuel-system, exhaust and heat-source locations are part of the G1 datum stack. The 48 V module domain is galvanically isolated from the host HV bus.

6.2 HV safety functions (hybrid HV battery and e-drive)

HV function	Baseline requirement
Battery enclosure	Protected, structural, serviceable, crash-contained; does not rely on the upper body for primary containment
HVIL	Continuous loop over pack, service disconnects, power electronics and charge interface
Isolation monitoring	Continuous during energized operation; fault response defined in ICD-001
Precharge	Controlled before inverter DC-link energisation
Crash isolation	Independent crash-signal path plus BMS / contactor logic
Service disconnect	Physically accessible and lockable
Thermal propagation	Cell-level detection, propagation-resistant pack design, vent routing
HV-to-MCRS separation	MCRS safe-state support remains available if traction HV is isolated
LV backup	Dedicated auxiliary energy sufficient for controlled safe-state functions

7. Fuel range screening

7.1 Assumptions

Tank-to-wheel efficiency η base = 0.22 (band 0.18–0.30) until supplier fuel maps arrive (ENE-H-001); the earlier 0.25–0.30 band was optimistic for a twin-turbo V8/V12 at very low load. Auxiliary load (HVAC, hydraulics, thermal) is included. Frontal area is scaled with the new heights, so CdA becomes 0.99 m² (V-L) and 1.6 m² (V-H): ASSUMPTION pending CFD.

Assumption	V-L	V-H	Note
Reference mass (curb + 200 kg)	2,978 kg	5,183 kg	Two occupants
CdA / Crr	0.99 m² / 0.009	1.6 m² / 0.012	CdA scaled with height
Tank-to-wheel η	0.18–0.30 (base 0.22)	0.18–0.30 (base 0.22)	ASSUMPTION; hybrid assist not credited
Auxiliary load	1 kW	2 kW	HVAC + hydraulics + thermal
Fuel energy density	8.9 kWh/L	8.9 kWh/L	Gasoline-class, ASSUMPTION
Fuel volume	80 L	150 L	Battery credit excluded

Method: wheel energy per km = (½·ρ·CdA·v² + Crr·m·g) ÷ 3.6; fuel use (L/100 km) = (wheel energy + P_aux/v) ÷ η ÷ 89; range = tank ÷ fuel use; ρ = 1.2 kg/m³.

7.2 Results (screening)

Case	Wheel Wh/km	L/100 km (η 0.30 / 0.22 / 0.18)	Range on fuel (η 0.30 / 0.22 / 0.18)
V-L 60 km/h	≈119	≈5.1 / 6.9 / 8.5	≈1,580 / 1,160 / 950 km
V-L 90 km/h	≈176	≈7.0 / 9.6 / 11.7	≈1,140 / 840 / 680 km
V-L 120 km/h	≈256	≈9.9 / 13.5 / 16.5	≈810 / 590 / 480 km
V-H 60 km/h	≈244	≈10.4 / 14.1 / 17.3	≈1,450 / 1,060 / 870 km
V-H 90 km/h	≈336	≈13.4 / 18.3 / 22.4	≈1,120 / 820 / 670 km
V-H 120 km/h	≈466	≈18.1 / 24.6 / 30.1	≈830 / 610 / 500 km

REQ-H01. The 1,000 km target (V-L) and 700–800 km target (V-H) are defined for a specified cycle (average speed ≤70 km/h for V-L). At η 0.22 the V-L 1,000 km limit is ≈74 km/h (≈85 km/h at η 0.25); the V-H 700 km target holds to ≈106 km/h and 800 km to ≈92 km/h. Monitoring at 90 and 120 km/h is sensitivity only. V-L does not reach 1,000 km at 120 km/h under any η in the band. Raising the height from 1,700 to 1,950 mm cost about 6 km/h of V-L cycle speed (≈80 → ≈74 km/h).

8. Controlled mass budget

Curb mass is the arithmetic sum of the groups below plus an explicit programme reserve. Design GVW = curb + occupants + payload (policy). All figures are TARGET / STUDY at evidence level L0. The height increase is assumed mass-neutral (OPEN).

Group	V-L (kg)	V-H (kg)	Note
Upper body (CFRP cell + roll + seats + closures)	755	1,220	Excl. occupants (policy mass listed below)
Battery (HV pack)	145	260	20 / 40 kWh usable class
Chassis, axles, wheels, brakes, steering	810	1,780	
ICE + hybrid drive + fuel + exhaust	580	820	≈20.9 % / 16.5 % of curb
MCRS module (four towers; Class S / L planning)	280	393	P50; see §9
HV/LV, cooling, protection	95	360	V-H includes optional 200 kg protection package
Reserve	113	150	≈4.1 % / 3.0 %
Curb total	2,778	4,983	TARGET
Occupants (policy)	400	600	100 kg each
Payload (policy)	100	400	
Design GVW	3,278	5,983	TARGET
Ceiling	4,200	7,000	LOCKED
Margin	965	1,017	DERIVED

Checks: V-L 755 + 145 + 810 + 580 + 280 + 95 + 113 = 2,778; 2,778 + 400 + 100 = 3,278. V-H 1,220 + 260 + 1,780 + 820 + 393 + 360 + 150 = 4,983; 4,983 + 600 + 400 = 5,983. If the optional 200 kg protection package is omitted it moves to reserve (identical treatment in the electric host).

P90 four-corner mass. Per-element deviations of the dedicated corner (7, 8, 4, 7, 3, 4, 4, 2 kg) give 14.9 kg RSS per corner; corners are assumed independent and the shared pump deviation (16 kg) is one further independent term: deviation = $\sqrt{(4 \times 223 + 16^2)} = 34$ kg, so P90 ≈ 427 kg. With the pump fully correlated (linear) P90 ≈ 439 kg; fully correlated arithmetic 4 × 125 + 48 + 17 = 565 kg. Excess over 393 kg: +34 / +46 / +172 kg. The 150 kg reserve covers both RSS cases only and is shared with every other group. Risk R-H2.

Class S exposure. The V-L module (280 kg) is a corner-only target (4 × 70) with no distribution allocation. If a Class S corner equals the Class L architecture, the module grows by 96 to 113 kg. The V-L reserve was raised from 70 to 113 kg (D-08) so that the full exposure is covered (curb 2,778 kg, GVW 3,278 kg, margin 922 kg). The design evidence is still OPEN (REQ-M30).

Difference to electric curb (R7.09): electric V-L ≈ +766 kg; electric V-H ≈ +712 kg (protection package treated consistently). Upper sprung mass (cockpit structure + occupants; payload rides in the trunk and does not load the module): V-L ≈ 1,155 kg; V-H ≈ 1,820 kg (without protection) or ≈ 2,020 kg (with protection).

9. Module mass, hydraulics and parameters

9.1 Corner assembly mass (Class L planning)

Element	P50 kg	P90 kg	Deviation (P90 − P50)	Scope
Local structure (per tower)	18	25	7	Core
Lorentz + local carriage	22	30	8	Core
Sensors + local electronics	8	12	4	Core
Core tower subtotal	48	67	RSS 11.4	Target ≤55 kg P50 met by estimate; limit until G2 = 67 kg
Cylinder + accumulator share	15	22	7	Corner
Valves / hoses / seals	5	8	3	Corner
Progressive stops	8	12	4	Corner
SPORT lock share	6	10	4	Corner
Harness / dampers / bellows	4	6	2	Corner
Pump + reservoir share (¼)	8	12	4 (16 for the shared pump)	Shared
Full corner assembly (Class L)	94	137		STUDY; ≈95 kg in host WBS

Core-tower RSS is 48 + √(7² + 8² + 4²) = 59.4 kg. The 67 kg arithmetic value stays the limit until G2 supplier correlation data; a 60 kg limit is accepted only with correlation evidence.

9.2 Module mass decomposition and P90 method

Authoritative decomposition for host mass tables: dedicated corner 86 / 125 kg (P50 / P90), shared pump 32 / 48 kg, distribution 17 kg (no P90 deviation assumed). Four-tower Class L P50 = 4 × 86 + 32 + 17 = 393 kg; six-tower P50 = 6 × 86 + 32 + 17 = 565 kg (one shared pump in both).

P90 method. Per-element deviations of the dedicated corner (7, 8, 4, 7, 3, 4, 4, 2 kg) combine to 14.9 kg RSS per corner. Corners are assumed independent: √n × 14.9. The shared pump deviation (16 kg) is one further independent term in the same quadrature sum, so the deviation is √(n × 223 + 16²). It is not added linearly.

Case	4-tower Class L	6-tower Class L	Meaning
P50	393 kg	565 kg	Budget value
P90, RSS (pump independent)	427 kg (+34)	605 kg (+40)	Method above; independence assumed
P90, RSS (pump fully correlated, linear)	439 kg (+46)	618 kg (+53)	Sensitivity; sensible upper RSS bound
P90, arithmetic (fully correlated)	565 kg (+172)	815 kg (+250)	Worst case: 4 × 125 + 48 + 17; 6 × 125 + 48 + 17

Reserve coverage: hybrid V-H reserve 150 kg covers both RSS cases but not the arithmetic case (+172 kg). Electric V-H (four towers, module 393 kg) reserve 220 kg covers all three cases, leaving 48 kg in the arithmetic case; the reserve is a programme reserve shared with every other group. Corner independence is itself an assumption until G2 supplier correlation data. Six towers (a module option, not baselined on either host) add ≈172 kg over four towers (565 − 393); the 605 / 618 / 815 kg figures apply to that option only.

9.3 Per-class corner mass and static support floor

Class	Mapped use	Corner P50	Module P50	Static basis per corner	Floor (1.25 × Fmax)
S	Light host (four seats)	≈70 kg (target)	≈280 kg (4 × 70, corner-only; distribution allocation OPEN)	Occupied share 310–350 kg	≥4.5 kN
M	Intermediate	≈85 kg	≈340 kg	Share 450–550 kg	≥7.0 kN
L	Heavy host	≈95 kg (94 incl. pump share)	393 kg (4 towers) / 565 kg (6 towers), incl. 17 kg distribution	Reference max 787.5 kg	≥10.0 kN

Class S exposure. The 70 kg corner has no design evidence. If a Class S corner equals the Class L architecture (94 kg), the four-tower module grows by 96 kg (corner-only) to 113 kg (with the 17 kg distribution). The hybrid V-L reserve (113 kg after D-08) and the electric V-L reserve (140 kg) both cover it; with the 275 kWh pack the electric V-L design GVW stays 43 kg inside the 4,200 kg ceiling. With payload in the trunk the V-L occupied share per corner is ≈295 kg ((780 + 400) ÷ 4), below the 310–350 kg basis; the 4.5 kN floor is kept as the conservative basis. OPEN: G1 Class S study (REQ-M30).

10. Module electrical and auxiliary energy

Module supply is 48 V class, galvanically isolated from host HV. Peak coil power per tower is $(F \div Km)^2 = (2{,}500\ N \div 63\ N/\sqrt{W})^2 \approx 1.575$ kW, so the micro-stage array peaks at ≈6.3 kW (four towers) and ≈9.45 kW (six towers). These are peak, not RMS, values and exceed the baseline DC-DC ratings of 3 kW and 4 kW. A local energy buffer (48 V battery or supercapacitor) is therefore mandatory for both configurations. The buffer serves comfort performance only; the S4 safety case does not rely on it. Safe-state energy reserve is verified under ENE-E-001 (electric) and ENE-H-001 (hybrid); power-loss detection and valve-closure timing are frozen in ICD-001.

Domain	Voltage	Isolation	Function
Module control / Lorentz	48 V class	Galvanic from host HV	Micro actuators, sensors, BCE
Hydraulic valves / hold	48 V / 12 V	Fail-closed on loss	Leveling, hold, lock actuation
Host traction HV	High voltage	Isolated from module	e-drive, HV battery (host responsibility)
Local energy buffer	48 V	Module domain	Peak shaving for Lorentz; mandatory in both configurations

10.2 Auxiliary energy budget (screening)

Auxiliary load for range screening includes HVAC, hydraulic pump (when active), thermal management and control electronics: 1 kW (V-L) and 2 kW (V-H) continuous. Peak Lorentz demand is covered by the local buffer, not by the continuous DC-DC rating. Buffer energy and recharge rate are sized in ENE-H-001 so that a worst-case isolation event sequence does not deplete the buffer below the safe-state reserve. On power loss the module retains passive hydraulic energy and mechanical capture without drawing on the buffer or host LV.

11. Control system, modes and state machine

11.1 Controller partition

Domain	Responsibility
Vehicle domain controller	Driver intent, drive state, dynamics, high-level requests; cannot bypass MCRS interlocks
MCRS Body Control ECU (BCE)	Mode arbitration, corner coordination, command bounds, lock sequencing, leveling, diagnostics
Independent safety monitor	Force / stroke / mode bounds, critical interlocks, watchdog and plausibility, where adopted
Corner controllers	Local sensing and actuator current / valve commands, bounded by BCE
BMS	Battery state, HV safety, contactor control, thermal limits
Brake controller	Primary service braking and regenerative coordination

Comfort control may be sophisticated; safety bounds remain simple, deterministic and independently monitorable. The monitor may be called independent only after power, clock, processor, software, communication and sensor dependencies are analysed and tested. OTA or calibration changes shall not bypass configuration control.

11.2 State machine

State	Name	Rule
S0	Safe / captured / power-off	No powered release; passive capture retained (equivalent to S4)
S1	Precharge / self-test	Capture healthy; auxiliary power path checked
S2	COMFORT	Macro + micro isolation active; capture mandatory
S2S	SPORT	Seat band; all required positive locks confirmed; stages held
S3	Degraded	Reduced authority; force and travel bounded
S4	Power-loss capture	Passive support; fail-safe valves; no software dependency for retention
S5	Impact preparation	High-rate unlock inhibited; stops prepared
S6	Impact / arrest	Progressive stops active; no actuator authority required
S7	Post-event safe	No automatic re-enable; manual clearance required
S8	Maintenance lockout	Physical lockout; stationary only

Any powered state goes to S4 on power loss (engaged locks stay engaged only if already locked; from COMFORT they stay released). S4 exits only to S1 after power restore and self-test. S2↔S2S only under the lateral-g policy with complete lock confirmation. The full transition table, including S5–S8 exit conditions, is required in ICD-001. Detection of power loss ($t_{det,max}$) must be hardware/passive so that S4 without software is not contradicted.

11.3 Mode sequences

Transition	Sequence
COMFORT → SPORT	Request accepted (no inhibit, lateral-g met, no fault) → macro drives to seat band → seat contact at required corners → locks engage → all required locks confirmed (incomplete set aborts to COMFORT) → stages held → S2S
SPORT → COMFORT	Allowed only under lateral-g policy → lock release commanded → all report released → macro envelope re-enabled → micro control re-enabled → S2
Power loss from COMFORT	Immediate S4; active force removed; passive support retained; locks stay released
Power loss from SPORT	S4 / fail-locked; locks remain engaged until controlled recovery
Critical fault	S3 / S4 as applicable; no automatic comfort restoration

12. Power-loss behaviour and lateral-g policy

12.1 REQ-M07 power-loss structure

After loss of commanded active power the system shall: (1) detect the loss within t_det,max (hardware/passive path); (2) enter S4; (3) close the relevant valves within t_valve,max; (4) remain inside the capture envelope; (5) stay below z_max and v_max; (6) prevent uncontrolled free-float; (7) settle in the defined S4 hydraulic state with passive support at or above the class static floor, with residual capacity and environment stated, under electrical loss and the defined hydraulic failure cases. State definition alone is not acceptance.

Limit to freeze	Where	Verified by
t_det,max, t_valve,max	ICD-001 (G2)	Bench + fault injection (L2/L3)
z_max, v_max, settle limit	ICD-001 (G2)	Single-tower power-cut test
Residual stiffness and damping in S4	ICD-001 (G1/G2)	L2 bench
Static support floor after hose-burst case	STR-001 (G2)	Hydraulic failure test
Energy reserve for safe-state functions	ENE-E-001 / ENE-H-001	Power-loss timing test

12.2 Lateral-g policy (REQ-M21)

Element	Rule
Inhibit	SPORT→COMFORT inhibited immediately when |ay| > 0.35 g, for automatic and manual requests
Re-enable	Only after |ay| stays below 0.30 g for the hold time
Driver override	May queue a request; cannot defeat the interlock
Filter and latency	Validated filter and measurement-to-inhibit latency budget, both biased to inhibit
Sensor faults	Validity, plausibility, disagreement thresholds, stale-data handling; a single lateral sensor cannot alone trigger an unsafe transition

Thresholds are temporary TARGET values; SIM-001 and HIL-001 freeze final thresholds, hysteresis, filtering and latency. PARAM-M21-ROLL-GRAD is defined at G2.

13. Structural principles

Structural screening loads are based on upper sprung mass (cockpit structure + occupants; payload rides in the trunk), not GVW. Corner loads assume equal share (÷4) and are a magnitude screen only. STR-001 must calculate real corner distribution including CG location, occupant and payload placement and load transfer under 1 g braking or cornering. Progressive stops provide the primary abnormal vertical energy path; capture frame and links provide the primary horizontal path; SPORT locks are not a substitute for stops.

13.1 Screening load cases (upper sprung mass)

Case	V-L (upper ≈1,155 kg)	V-H (upper ≈2,020 kg incl. protection)	Note
1 g static, per corner	≈2.8 kN	≈5.0 kN	Cockpit + occupants (payload in trunk)
2 g vertical	≈5.7 kN	≈9.9 kN	Screening
20 g crash vertical (four towers)	≈57 kN	≈99 kN	Stops path
1 g braking transfer	TBD	TBD	STR-001 — not equal share
Cornering transfer	TBD	TBD	STR-001 — CG dependent

The six-tower option would lower the V-H 20 g corner load to ≈66 kN at a cost of ≈172 kg; it is not baselined (D-02). Payload rides in the trunk (§3 body zoning) and does not load the module.

13.2 Progressive stop envelope

Stops shall absorb the design abnormal vertical energy without reliance on the Lorentz actuator or closed-loop force. Force–displacement curve, residual height and rebound are frozen in STR-001. Stop stroke is only a 50 mm placeholder in the packaging stack (OPEN, G1).

14. Requirements (selected hybrid-host)

The full requirements register lives in the Master and ICD-001. Selected hybrid-host items:

ID	Statement (summary)	Status
REQ-H01	Range target 1,000 km (V-L) / 700–800 km (V-H) defined for cycle average ≤70 km/h (V-L); 90 / 120 km/h are sensitivity only	OPEN — ENE-H-001
REQ-H06	V-H tower count: four Class L towers (decided 2026-10-12); six-tower option not baselined	BASELINE
REQ-M07	Power-loss structure (detection, valves, envelope, settle)	OPEN — ICD-001
REQ-M17/M20	Lock state on power loss: engage only if already locked; COMFORT stays released	OPEN — LCK-001
REQ-M21	Lateral-g interlock 0.35 / 0.30 g	TARGET (temporary)
REQ-M30	Class S corner mass basis including distribution allocation	OPEN — G1
REQ-M31	Body zoning: only the safety cockpit is suspended; battery, powertrain, fuel, hood and trunk payload on the lower chassis; seam envelope in ICD-001	BASELINE / seam OPEN — ICD-001
REG-001	Regulatory exposure incl. V-L ceiling 4,200 kg above the 3,500 kg threshold in some markets	OPEN — G1
REQ-H07	Seam firewall (hybrid): heat, exhaust gas and fuel vapour shall not reach the cell through the seams; ±57.5 mm normal motion and 107.5 mm overtravel (Master M.10)	OPEN — ICD-001 / THERM-001
REQ-H08	Class S reserve: the V-L hybrid reserve (113 kg) covers the Class S exposure (+113 kg); evidence in the G1 Class S study	BASELINE (D-08)
REQ-M32 to M36	Packaging margin ≥ +20 mm, vertical arrest (η ≥ 0.70, 3.0 m/s baseline), horizontal crash path, seam function, ceiling sizing (Master Appendix M)	OPEN — Master App. M

15. Safety analysis

Screening FMEA exists at L0. No HARA / FTA or independence argument yet. Closure vehicle: SAF-001 (G2). High-severity (S ≥ 9) items require explicit pre-G2 review independent of RPN; RPN scoring does not hide high-severity rows.

15.1 Screening FMEA summary (selected rows)

ID	Failure mode	Effect	S	O	D	RPN	Action
F-01	Loss of geometric capture	Upper body free	10	2	4	80	Pre-G2 review; redundant paths
F-02	Link / carriage fracture	Horizontal path lost	9	2	5	90	Pre-G2 review; STR-001
F-03	SPORT lock incomplete set	False SPORT declaration	8	3	3	72	Per-lock confirmation logic
F-04	Power-loss valve fails open	Loss of passive support	9	2	4	72	Fail-closed design; test
F-05	HV fault propagates to module	Capture defeated	9	2	5	90	Galvanic isolation; HVIL
F-06	Fuel / heat near cell	Thermal event	8	3	4	96	THERM-001; packaging
F-07	Single lateral sensor fault	Unsafe mode transition	7	3	4	84	Plausibility + dual path
F-08	Software command out of bounds	Force exceedance	7	3	3	63	Independent monitor

Severity ≥9 rows are flagged for mandatory pre-G2 independent review regardless of RPN. Full FMEA is in SAF-001.

16. Verification

Current position L0 (architecture). SIM-001 (G1) and the tower bench (G2) are the first simulation and hardware vehicles. All state-machine transitions require HIL evidence before credit.

Level	Meaning	Typical artefacts
L0	Architecture definition, arithmetic consistency	This baseline
L1	First-order simulation / screening	SIM-001
L2	Component / subsystem bench	Tower bench, ACT-001
L3	HIL / system integration	State-machine HIL, power-loss timing
L4	Vehicle-level prototype	Instrumented mule
L5	Production-intent / certification	Beyond this baseline

Gate	Focus	Key exit criteria
G0	Document release	Owner release; reviewer appointed
G1	Architecture freeze / packaging	3-D datum stack validating the re-baselined stack with residual ≥ +20 mm or re-baselined heights (REQ-M32); design vertical impact speed fixed (REQ-M33); ICD-001 draft; REG-001; cycle definition; Class S study (REQ-M30); V-L pack rule applied to supplier data (REQ-M36 / REQ-E03)
G2	Subsystem evidence	Tower bench; SAF-001; supplier data; ACT / STR / ENE closure; stop-curve efficiency measured (REQ-M33); horizontal path sized (REQ-M34); seam function, hybrid firewall and electric vent isolation demonstrated (REQ-M35, REQ-H07, REQ-E02)
G3	System HIL	Full state machine; power-loss; lateral-g
G4	Vehicle mule	Instrumented range and structural tests
G5	Production readiness	Beyond this baseline

17. Gates, risks and open items

17.1 Open items (selected)

#	Item	Closure
1	Packaging re-baselined stack (0 / +10 mm) conditional on 170 mm of unvalidated levers (P0)	G1 3-D datum stack
2	P90 module mass (RSS vs arithmetic); supplier correlation	G2 supplier data
3	Fuel maps and η confirmation	ENE-H-001
4	Height sensitivity: mass, CG and CdA effect of +250 / +130 mm	G1 SIM / CFD
5	Lock / S4 logic from COMFORT	ICD-001 / LCK-001
6	Full state-machine transition table S5–S8	ICD-001
7	Regulatory 3,500 kg threshold (ceiling 4,200 kg)	REG-001 G1
8	IP filing receipt	Counsel
9	Independent technical reviewer appointment	≤2026-12-05 or G1
10	Energy buffer sizing (4- and 6-tower)	ENE-H-001 / ICD-001
11	Stop stroke and micro-stage clearance (50 mm placeholder)	STR-001 / G1
12	Structural corner load distribution (CG, transfer)	STR-001
13	High-severity (S ≥ 9) pre-G2 review	SAF-001
14	PARAM-M21-ROLL-GRAD definition	G2
15	Protection package inclusion policy (both hosts)	G1 mass closure
16	Class S distribution allocation and mass evidence (REQ-M30); V-L reserve 113 kg (D-08)	G1 Class S study
17	Lever feasibility: free play 10 mm, 60 mm shared tower depth	ICD-001 / STR-001
18	Seam envelope between the moving cell and fixed bodywork (water, dust, noise, pinch)	G1 / ICD-001
19	Lower bodywork (hood, wings, trunk, rear body) mass itemisation; currently inside the chassis group (A-15)	G1 mass freeze
20	Design vertical impact speed (baseline 3.0 m/s) and stop-curve efficiency η ≥ 0.70 (REQ-M33)	STR-001 / G1
21	Horizontal crash path: capture-frame load bound at 20 g (REQ-M34)	STR-001 / SAF-001
22	Seam firewall to engine-bay heat, exhaust gas and fuel vapour (REQ-H07)	ICD-001 / THERM-001

17.2 Risk register (selected)

ID	Risk	L	I	Score	Mitigation
R-22	Re-baselined stack closes only if all 170 mm of levers are delivered	4	4	16	G1 3-D datum stack
R-H2	Four-corner P90 mass exceeds the 150 kg reserve (arithmetic case)	4	4	16	G2 supplier correlation
R-H3	Fuel range claim fails at higher speed	4	3	12	ENE-H-001; cycle ≤70 km/h
R-H4	ICE heat / exhaust near cell	3	4	12	THERM-001; packaging
R-H5	Regulatory 3,500 kg threshold	3	3	9	REG-001; market selection
R-H6	Lock logic ambiguity on power loss	3	4	12	ICD-001 / LCK-001
R-H7	No filing receipt (IP)	3	3	9	Counsel; provisional
R-H8	Energy buffer undersized	3	3	9	ENE-H-001 sizing
R-H9	Class S module +113 kg covered by the 113 kg V-L reserve (D-08); design evidence required	2	3	6	G1 Class S study
R-H10	Height increase raises CdA and mass; range targets move	3	3	9	G1 SIM / CFD
R-H11	Seams between the moving cell and fixed bodywork not designed (water, dust, noise, pinch)	3	3	9	ICD-001 / G1
R-H12	Stop-curve efficiency below 0.70 or design impact speed above 3.7 m/s: the 50 mm stroke is not enough and the stack grows	3	4	12	STR-001
R-H13	Horizontal crash path into the cockpit unsized (≈57 / 99 kN per link at 20 g)	3	4	12	STR-001 / SAF-001

18. Controlled document tree

Master MCRS-DOC-00-R8.09 (module values) → Hybrid MCRS-DOC-00-R5.40 (this file) and Electric MCRS-FE-DOC-00-R7.09. This document is the controlling hybrid-host Document 00 under the Master; it does not control module values.

Document / WP	Role	Owner / gate
MCRS-DOC-00-R8.09	Master — module values	Owner; governs module
MCRS-DOC-00-R5.40	Hybrid host baseline (this)	Owner; hybrid values
MCRS-FE-DOC-00-R7.09	Electric host baseline	Owner; electric values
ICD-001	Interface control document	G1 draft
SIM-001	System simulation	G1
ENE-H-001	Hybrid energy / fuel maps	G1/G2
SAF-001	Safety analysis / HARA	G2
STR-001	Structural / stops	G2
LCK-001	Lock logic and confirmation	G1/G2
REG-001	Regulatory exposure	G1
THERM-001	Thermal / exhaust packaging	G2
ACT-001	Actuator characterisation	G2

19. IP, disclosure and commercial paths

Founder / Legal Owner: Amirbahador Atai (individual rights holder). Assignment to a legal entity is a separate controlled act. Licensable module: MCRS-MOD-001. Paths A / B / C as in §1.2: Path A is a paid Option covering the SIM-001 run and ICD-001 draft (buyer may bring its own host; no vehicle sale); Path B is host engagement studies; Path C is IP-transaction discussion under counsel only (T3). Wording: integration into a two-piece vehicle body (not “EV body”). Trade-mark reference: MCRS. No filing receipt on record; provisional filing under counsel is OPEN. This document is T1 Confidential / NDA controlled; distribution is limited to parties under valid NDA; public disclosure beyond the claims-discipline table is prohibited until filing and owner release.

20. Claims discipline and glossary

Claim type	Allowed statement	Forbidden statement
Range (electric)	“≥800 km design capability under defined cycle (STUDY)”	“800 km verified / certified range”
Range (hybrid)	“1,000 km target for cycle ≤70 km/h avg (STUDY)”	“1,000 km verified range at any speed”
Energy	“275 kWh (V-L) / 300 kWh (V-H) usable TARGET”	“certified capacity”
Safety	“Invariants defined; screening FMEA at L0”	“ASIL-rated”, “crash-certified”
Comfort	“Targeted isolation (unproven at L0)”	“Superior ride”, clinical outcome
Mass	“P50 closed with reserve; P90 OPEN”	“Mass closed”, “production mass”
Packaging	“FIXED-FOR-ENGAGEMENT / OPEN-P0”	“Locked geometry”, “packaging closed”
Readiness	“TRL 2–3, evidence L0”	“Production ready”, “homologated”
Actuator	“Km ≥63 N/√W ASSUMPTION”	“Proven force capability”

Term	Definition
CMFSB	Captured Mechanical Floating Safety Body — the isolated upper safety cell architecture
MCRS / MCRS-MOD-001	Modular Captured Ride System (product family) / the licensable four- or six-tower module
COMFORT (S2) / SPORT (S2S) / S4	Isolation mode / locked mode with all required locks confirmed / power-loss capture
Class S / M / L	Module corner mass and load class (light / intermediate / heavy host)
P50 / P90 / RSS	Percentile mass estimates; root-sum-square combination of independent deviations
FIXED-FOR-ENGAGEMENT / OPEN-P0	Locked for packaging studies but not closed / blocking packaging item
L0–L5 / G0–G5	Evidence ladder levels / programme gates
BCE / HVIL	MCRS Body Control ECU / High-Voltage Interlock Loop
Usable energy	Net energy available after the reserved / inaccessible fraction (≈10 %)
RA-1	Ambulance / specialty medical transport (comfort context only; no clinical claim)

Appendix A — Correction log

A.1 R5.34 → R5.35 (retained)

#	Item	Action in R5.35
HA-01	IDs shared by Corrected and Fixed passes; Master reference R8.02	New ID R5.35; cites Master R8.04
HA-02	P90 text “shared pump linear” did not reproduce 427 / 605 kg	Method stated exactly in §8 (pump = one independent term); linear-pump sensitivity 439 kg added
HA-04	Class S distribution allocation not stated	§8 states corner-only 280 kg and the +96 to +113 kg exposure
D-01	Owner decision: V-L ceiling 4,200 kg	§2, §5, §8 updated; V-L margin 965 kg at that issue (922 kg since D-08)
D-02	Owner decision: four V-H towers on both hosts	Tower option removed from baseline; REQ-H06 BASELINE
D-03	Owner decision: heights 1,950 / 2,130 mm	Stack re-baselined (0 / +10 mm, conditional); CdA scaled; fuel range recomputed; V-L cycle ≤70 km/h
CL-09	Range text used the 120 km/h V-L case loosely	REQ-H01 restated with limits (≈74 km/h at η 0.22)

Earlier closures (R5.34) remain closed: energy buffer for both configurations, packaging module height ≥260 / ≥280 mm, lock / S4 logic, status tags, S ≥ 9 review independent of RPN, MCRS-MOD-001 uniform ID, risk R-H2, IP wording, protection-package consistency.

A.2 R5.35 → R5.36 (retained)

#	Item	Action in R5.36
D-05	Owner decision 2026-10-12: only about half of the body (the passenger cell) is suspended; battery and powertrain are in the lower chassis and add no weight on the upper level	§3 mirrors Master §3.4: zones, side-view figure, seam envelope, payload-location rule; REQ-M31; open item 18; risk R-H11
CL-11	Battery relationship unclear	Stated: the battery loads neither the cell nor the module; it counts only in vehicle weight vs ceiling and, by thickness, in chassis stack height
CL-12	Payload implicitly assumed in the cell	Conservative baseline kept; payload in the rear lower zone is a sensitivity (V-H upper mass ≈2,020 kg, 20 g corner ≈99 kN)

A.3 R5.36 → R5.37 (retained)

#	Item	Action in R5.37
D-07	Owner decision 2026-10-12: only the safety cockpit is suspended; hood and trunk are attached to the lower body; payload rides in the trunk	§3 mirrors the rewritten Master §3.4; §8 and §13.1 recomputed: V-L upper ≈1,155 kg, V-H ≈2,020 kg; 20 g corner ≈57 / 99 kN; six-tower option ≈66 kN
D-06	Owner decision: electric V-L battery 275 kWh (V-H electric stays 300 kWh)	§2.3 comparison and §8 electric curb difference updated (V-L electric curb 3,544 kg, GVW 4,044 kg, margin +156 kg)
CL-13	Lower bodywork mass not itemised	Assumption A-15 in Master and Electric: carried inside the chassis group; open item 19

A.4 R5.37 → R5.38 (retained)

#	Item	Where recorded
EC-01, EC-05, EC-07, EC-09, EC-10	Consistency defects found in R5.37	Appendix D.1
EN-01 to EN-06	Engineering findings of the R5.38 review	Appendix D.2

A.5 R5.38 → R5.39 (retained)

#	Item	Action in R5.39
D-08	V-L hybrid reserve raised from 70 to 113 kg to cover the Class S exposure	Curb 2,778 kg; design GVW 3,278 kg; margin 922 kg; ICE share 20.9 %; fuel reference mass 2,978 kg; REQ-H08 BASELINE; R-H9 reduced
RV-01 to RV-12	Review findings of R5.40	Appendix D.3; tables D.5 to D.7
Page rule	New page count at least the previous page count	R5.39 has 27 pages against 25 in R5.38

A.6 R5.39 → R5.40 (this issue)

#	Item	Action in R5.40
RX-01 to RX-06	Review findings of R5.40	Appendix D.8; Appendix E
Page rule	New page count at least the previous page count	R5.40 has 30 pages against 27 in R5.39

Appendix B — Status tag definitions (normative)

Tag	Meaning	Change control
BASELINE	Architecture decision frozen for this issue	Requires revision and owner release
TARGET	Design intent under evidence	G1/G2 evidence can refine
STUDY	Screening only; not design commitment	Superseded by supplier / test data
FIXED-FOR-ENGAGEMENT	Locked for packaging studies; not packaging-closed	Re-baseline at G1 if stack fails
DERIVED	Arithmetic consequence of controlled values	Recalculated when inputs change
OPEN — P0	Blocking packaging open item	Must close or re-baseline before G1 exit
ASSUMPTION	Explicit assumption with owner and closure route	Closed by named work package
LOCKED	Hard ceiling or constraint	Programme policy change required
LOCKED–EXCEEDED	A LOCKED value that a controlled design value breaches	Open decision at the named gate; the breach must be shown wherever the value appears

Previous use of LOCKED for open packaging heights was misleading and is replaced by FIXED-FOR-ENGAGEMENT throughout.

Appendix C — Post-issue correction record (C-series)

- C-01 (R5.35): heights, ceiling and tower count updated per owner decisions D-01 to D-03 of 2026-10-12.
- C-02 (R5.35): fuel-range table recomputed with CdA 0.99 / 1.60 m²; V-L 1,000 km limit ≈74 km/h at η 0.22 at that issue (reference mass 2,935 kg; ≈74 km/h with 2,978 kg since D-08, target cycle ≤70 km/h); V-H 700–800 km holds to ≈106 / ≈92 km/h.
- C-03 (R5.35): P90 method wording corrected; Class S exposure disclosed; references updated to Master R8.04 and Electric R7.04.
- C-04 (R5.34, carried): mass budget re-summed (V-L 2,735 kg at that issue, 2,778 kg since D-08; V-H 4,983 kg); four-tower Class L module 393 kg; regulatory note — V-L hybrid design GVW (3,235 kg at that issue, 3,278 kg since D-08) is below 3,500 kg.
- C-05 (R5.36): body zoning and load boundary added (§3 mirror of Master §3.4); payload-location sensitivity added (superseded by D-07); references updated to Master R8.05 and Electric R7.05.
- C-06 (R5.37): cockpit-only suspension with hood, trunk and payload on the lower body; upper sprung mass and corner loads recomputed without payload; electric V-L comparison for the 275 kWh pack; references updated to Master R8.06 and Electric R7.06.
- C-07 (R5.38): review of R5.37 (EC-01, EC-05, EC-07, EC-09, EC-10; EN-01 to EN-06); REQ-H07 and REQ-H08 added; references updated to Master R8.07 and Electric R7.07.
- C-08 (R5.39): review RV-01 to RV-12 applied; D-08 V-L reserve 113 kg; fuel table recomputed with reference mass 2,978 kg; stop-stroke efficiency and design speed, horizontal capture bound, seam envelope and height sensitivity added (Appendix D); R5.39 has 27 pages against 25 in R5.38; references updated to Master R8.08 and Electric R7.08.
- C-09 (R5.40): review RX-01 to RX-06 applied: G1 / G2 exit criteria include REQ-H07, REQ-H08 and REQ-M32 to REQ-M36 (§16); Appendix E added (release readiness and contract use); R5.40 has 30 pages against 27 in R5.39; references updated to Master R8.09 and Electric R7.09.

Appendix D — Design and engineering review (R5.40)

This appendix records the review history of this document. Review R5.38 found consistency defects in R5.37 and engineering findings; review R5.39 found further defects in R5.38; review R5.40 found the gap described in D.8 (RX-01). Calculations are first-order (L0); they set bounds and closure criteria and do not replace evidence from SIM-001, STR-001, ICD-001 or supplier data. Module-level requirements REQ-M32 to REQ-M36 are defined in Master R8.09 Appendix M.

D.1 Consistency defects closed in R5.38 (retained)

ID	Finding in R5.37	Resolution in R5.38
EC-01	Master R8.06 §9.2 stated that every V-L design GVW is above 3,500 kg; the hybrid V-L design GVW is below it.	Corrected in the Master; this document already stated the hybrid design GVW below the threshold and the 4,200 kg ceiling above it (REG-001).
EC-05	The body-zoning table listed the electric battery as 300 kWh; V-L electric is 275 kWh.	Corrected to electric 275 / 300 kWh (§3.4).
EC-07	The C-series record was not in ID order and Appendix A.3 was tagged as this issue.	C-series in ID order; A.3 tagged retained.
EC-09	New content was issued under the existing ID R5.37.	Issued under a new ID; revision history, document tree and supersession updated.
EC-10	Page count must not fall below the original issue.	R5.38 had 25 pages against 24; restated as the page rule for R5.40.

D.2 Design and engineering findings (hybrid host) and status in R5.40

ID	Finding	Resolution / requirement	Status in R5.40
EN-01	The stack closes at 0 mm (V-L) and +10 mm (V-H) with every lever unvalidated.	REQ-M32: residual ≥ +20 mm at G1 or re-baseline (V-L ≈1,970 mm, V-H ≈2,140 mm).	OPEN; consequences in Master M.9 and table D.7
EN-02	The 50 mm stop stroke has no link to a design impact speed.	REQ-M33.	Refined by RV-01 (table D.5)
EN-03	The horizontal crash path into the cockpit is not sized.	REQ-M34.	Bound in table D.6; OPEN
EN-04	The front seam faces engine-bay heat, exhaust gas and fuel vapour.	REQ-M35 and REQ-H07.	OPEN — ICD-001 / THERM-001
EN-05	Average coil power was not stated.	$(700 \div 63)^2 \approx 123$ W per tower, ≈0.49 kW for four towers, inside the 3 kW DC-DC.	Closed
EN-06	The V-L hybrid reserve (70 kg) was below the Class S exposure (+113 kg).	Option D-08.	Closed by D-08 (RV-02)

D.3 Review R5.39: findings in R5.38 and resolutions

ID	Finding in R5.38	Resolution in R5.39
RV-01	The arrest table assumed an ideal constant-force stop.	Efficiency η = 0.70 and baseline design speed 3.0 m/s: stroke 33 mm against 50 mm (table D.5).
RV-02	D-08 was an unapplied option; R-H9 stayed open although the margin is 965 kg.	D-08 applied: reserve 113 kg, curb 2,778 kg, GVW 3,278 kg, margin 922 kg; REQ-H08 BASELINE.
RV-06	The horizontal capture-frame load was not bounded.	F = m·a at 20 g and 30 g (table D.6).
RV-07	Height and CdA effects on the hybrid range were not quantified.	Table D.7.
RV-09	The seam had no overtravel requirement.	107.5 mm overtravel; firewall function (REQ-H07).
RV-11	The page rule was “must not fall below”.	R5.39 has 27 pages against 25 in R5.38.

D.4 Hybrid-host requirement additions

ID	Statement	Status
REQ-H07	Seam firewall (hybrid): heat, exhaust gas and fuel vapour shall not reach the cell through the seams.	OPEN — ICD-001 / THERM-001
REQ-H08	Class S reserve: V-L reserve 113 kg covers the Class S exposure (+113 kg); evidence by the G1 Class S study.	BASELINE (D-08)
REQ-M32 to M36	Packaging margin, vertical arrest, horizontal crash path, seam function, ceiling sizing (Master Appendix M).	OPEN — Master App. M

D.5 Vertical arrest (η = 0.70, 20 g mean deceleration)

Design impact speed	Stroke ideal (η = 1)	Stroke at η = 0.70	Energy per corner V-L / V-H	Consequence
3.0 m/s (baseline)	23 mm	33 mm	1.30 / 2.27 kJ	Fits 50 mm with 17 mm spare
3.7 m/s	35 mm	50 mm	—	Limit of the 50 mm placeholder
4.4 m/s	50 mm	72 mm	—	Needs +22 mm
6.0 m/s	92 mm	131 mm	—	Needs +81 mm; stack re-baseline

D.6 Horizontal capture-frame load bound

Host	Upper sprung mass	Force at 20 g	Per link (four)	Force at 30 g
V-L hybrid	1,155 kg	227 kN	57 kN	340 kN
V-H hybrid	2,020 kg	396 kN	99 kN	594 kN

Equal share over four links; the 20 g mean pulse is a planning assumption for STR-001 and SAF-001 to replace.

D.7 Hybrid range sensitivity to height (V-L cycle limit for 1,000 km at η 0.22)

Case	Height	CdA	V-L cycle limit
Baseline	1,950 mm	0.99 m²	≈74 km/h
+10 mm	1,960 mm	+0.51 %	≈74 km/h
+20 mm (REQ-M32 margin)	1,970 mm	+1.03 %	≈73 km/h
All levers fail (+170 mm)	2,120 mm	+8.7 %	≈70 km/h

D.8 Review R5.40: finding in R5.39 and resolutions

ID	Finding in R5.39	Resolution in R5.40
RX-01	The G1 and G2 exit criteria (§16) did not include REQ-H07, REQ-H08 and REQ-M32 to REQ-M36.	§16 mirrors the extended Master criteria: G1 adds REQ-M30, M32, M33, M36; G2 adds REQ-M33 (stop-curve efficiency measured), M34, M35 and REQ-H07 (hybrid firewall).
RX-02 to RX-05	Assumptions, contract use and release conditions were not collected.	Master Appendix N and Appendix E of this document.
RX-06	Page rule.	R5.40 has 30 pages against 27 in R5.39.

Appendix E — Release readiness and contract use (hybrid host)

The general statement is in Master R8.09 Appendix N: this document may be used for NDA diligence, Path A Option scoping, host engagement studies and IP discussion under counsel; it must not be used for any warranty of range, safety, crashworthiness, homologation, mass or packaging, or for production or price commitments.

Hybrid-host item	Status	Closure
Owner release and confirmation of the register (Master N.1)	Pending	Owner
D-08: V-L reserve 113 kg (curb 2,778 kg)	Applied on the owner's instruction; confirmation pending	G1 mass freeze
Fuel range: η band, CdA scaled with height, cycle ≤70 km/h	Screening only; no fuel maps	ENE-H-001
Hybrid firewall to engine-bay heat, exhaust gas and fuel vapour (REQ-H07)	Open	ICD-001 / THERM-001
Stack closure (0 / +10 mm, conditional)	Open — P0	G1 3-D datum stack
Independent Technical Reviewer and counsel review	Not appointed	Owner

Role	Name	Date	Signature
Owner release (Legal Owner)	Amirbahador Atai	To be completed	To be completed
Independent Technical Reviewer	To be appointed	To be completed	To be completed

— END OF MCRS-DOC-00-R5.40 —